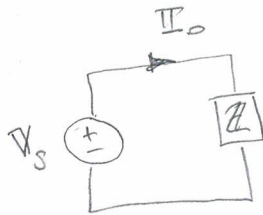


7.13)



SSK MAXWELT, P .
REAKTIVWELT Q
KOMPLEX WELT Φ

$$a) \begin{cases} v_s(t) = 650 \cos(377t) \quad \checkmark \\ i_o(t) = 20 \cos(377t - 10^\circ) \text{ A} \end{cases}$$

Winkel: $\phi_v - \phi_i = 0 - (-10^\circ) = 10^\circ$

$$P_f = \cos \phi = \cos(\phi_v - \phi_i) = \cos(10^\circ) \approx 0.9848$$

$$\Phi = \frac{1}{2} V I^* = \frac{1}{2} 650 \angle 0^\circ 20 \angle 10^\circ \approx 6500 \angle 10^\circ \text{ VA}$$

$$\begin{cases} P = |\Phi| \cos(10^\circ) = 6500 \cos(10^\circ) \approx 6401.3 \text{ W} \\ Q = |\Phi| \sin(10^\circ) = 6500 \sin(10^\circ) \approx 1128.7 \text{ VAR} \end{cases}$$

$$b) \Phi = V_{\text{rms}} I_{\text{rms}}^* = 460 \angle 0^\circ 14.14 \angle 45^\circ = 6504 \angle 45^\circ \text{ VA}$$

$$\begin{cases} P = |\Phi| \cos(45^\circ) = 6504.4 \cos(45^\circ) \approx 4599.3 \text{ W} \\ Q = |\Phi| \sin(45^\circ) = 6504.4 \sin(45^\circ) \approx 4599.3 \text{ VAR} \end{cases}$$

$$c) \Phi = V_{\text{rms}} I_{\text{rms}}^* = 150 \angle 0^\circ 8.6 \angle +86^\circ = 860 \angle 86^\circ \text{ VA}$$

$$\begin{cases} P = |\Phi| \cos(86^\circ) = 860 \cos(86^\circ) \approx 51.97 \text{ W} \\ Q = |\Phi| \sin(86^\circ) = 860 \sin(86^\circ) \approx 857.91 \text{ VAR} \end{cases}$$

$$d) \Phi = V_{\text{rms}} I_{\text{rms}}^* = 208 \angle -30^\circ 7.3 \angle +63^\circ = 478.4 \angle +33^\circ \text{ VA}$$

$$\begin{cases} P = |\Phi| \cos(+33^\circ) = 478.4 \cos(+33^\circ) = 401.22 \text{ W} \\ Q = |\Phi| \sin(+33^\circ) = 478.4 \sin(+33^\circ) = 260.56 \text{ VAR} \end{cases}$$

SWAR: P & Q & W